Sahaj Bhadja

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Summary

Software Engineer with 2+ years in **QA-oriented delivery**, creating **manual and automated validation workflows**, logging defects, and improving release quality for cloud software using Python, Java, SQL, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Developed data workflows across **Azure SQL/PostgreSQL** for 500+ users, reducing dashboard latency from 1.2s to 350ms through query tuning and schema improvements.
- Designed release validation and reconciliation checks for 5K+ monthly jobs, logging issues early and improving software readiness for downstream business and client acceptance testing.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Built and tested Flask + React features handling 10K+ daily requests, converting user stories into modular APIs and workflow checks aligned with iterative Agile delivery.
- Implemented concurrent-processing safeguards and CI/CD controls via GitHub Actions, reducing turnaround time by 60% while maintaining stable behavior during peak instructor-facing usage periods.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Created Python experimentation pipelines processing 100K+ trajectory logs with reproducible runs, structured outputs, and validation tracking to support reliable analysis and defect reproduction workflows.
- Documented test outcomes, debugging traces, and issue triage notes, improving collaboration with researchers and accelerating fix verification across iterative simulation and model-testing cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Developed and maintained 20+ **unit and integration test suites** with Jenkins, PyTest, and Mockito across three OS environments, strengthening regression confidence and release quality gates.
- Troubleshot API and ETL defects using OOP, AWS Glue, and S3, improving response time by 30%, increasing pipeline efficiency by 68%, and reducing technical debt.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built REST API and WebSocket modules supporting eight workflows and 2M+ synchronized records, adding operational checks and logs to simplify debugging and failure diagnosis.
- Automated deployment and testing with Jenkins and Docker, reducing deployment time by 50% and improving consistency across release candidates, hot-fix cycles, and environment migrations.

Skills

Languages: Java, **Python**, SQL, JavaScript/TypeScript, C++
QA & Release: Manual testing, regression testing, defect logging, fix verification, release validation, CI/CD
Automation: Jenkins, GitHub Actions, PyTest, Mockito, Selenium familiarity, test case development
Cloud & Platforms: Azure cloud workflow, Linux/Unix command-line workflow, Docker, Kubernetes, Git
Backend & Data: Flask, FastAPI, Node.js, REST APIs, PostgreSQL, MySQL, MongoDB, Azure SQL
Process: Agile/Scrum, SDLC, technical documentation, cross-functional collaboration

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built a multi-database SQL workspace supporting 10+ databases with schema-aware query generation, execution validation, and structured summaries, improving reproducibility and reducing manual errors during test scenarios.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed API-driven healthcare workflows with monitoring-backed checks, supporting 100+ daily appointments while improving record retrieval by 2x and strengthening end-to-end reliability across operational and user workflows.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented six scheduling algorithms and interactive visual controls across 1K+ process combinations, improving debugging clarity, algorithm verification, and documentation of expected versus observed execution behavior.